

SOLVING PROBLEMS USING PROBABILITY



LESSON INTRODUCTION

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

LEARNING TARGETS

- I can use theoretical probability to make predictions.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.1 Determine the sample space for a simple experiment.

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

WORKING TOWARD

MA.912.DP.4.1 Given a mathematical or real-world situation, calculate the appropriate permutation or combination.

MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.

MA.912.DP.3.5 Solve real-world problems involving univariate and bivariate categorical data.

ESSENTIAL QUESTIONS

- How can probabilities be used to make predictions?

LESSON NARRATIVE

In this lesson, students apply their knowledge of finding probability to Punnett Squares. Students create Punnett Squares to determine the probability that puppies and kittens will be born with certain genetic traits. Students then make predictions to determine numbers of sample spaces and populations based on given numbers and probability. This lesson offers a nice connection to science. Students learn to determine the probabilities for genotypes and phenotype combinations using Punnett Squares in Grade 7 science. Revisiting this concept here in its relationship to theoretical probability creates a cross-curricular connection that highlights the utility of mathematics in multiple fields.

COMPONENT SUMMARY

14.7.1 WARM-UP (~5 MIN.)

Determine possible combinations

14.7.3 YOUR TURN! (10 MIN.)

Make predictions from real-life probability models

14.7.2 COLLABORATION (20-25 MIN.)

Make predictions and find total outcomes

14.7.4 WRAP-UP (~5 MIN.)

Make a prediction from a real-world situation

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Prediction
- Likelihood

MATERIALS

- (Optional) Chart paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with representing probabilities in different forms and converting between them.

THINGS TO CONSIDER

- Consider collaborating with a science teacher before implementing this lesson. Students work with Punnett Squares in Grade 7 science. This lesson provides an interdisciplinary focus.

LITERACY AND LANGUAGE ROUTINES

Take Turns

Pair 14.7.2 Collaboration with a Take Turns routine to encourage students to engage in meaningful mathematical discourse throughout this section. Encouraging students to explain and justify their responses will deepen understanding as students make connections between what they're learning about probability and its application to other areas, such as science.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

This lesson encourages cross-curricular connections through the use of Punnett Squares to teach and practice probability. One of the clarifications of this standard includes demonstrating how various concepts can be applied to other disciplines, which is something students are able to do throughout this lesson.

DIFFERENTIATE INSTRUCTION

Consider providing Desmos Activity “Chance Experiments” as support for remediating student conceptual understanding of finding probability. This digital interactive provides additional practice as students work through probability and make sense of its application in the real world.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

Let's play a game.

Here's how it works. You'll spin the spinner.

How do you win? That's up to you. Two options:

(1) Spinner lands on RED, you win.
(2) Spinner lands on BLUE, you win.

Which option do you want?

14.7.1 WARM-UP: MEAL COMBINATIONS

14.7.1 Warm-Up: Meal Combinations

- How many different meals are possible if each meal includes one main course, one side dish, and one drink?

Main Courses	Side Dishes	Drinks
grilled chicken	salad	milk
pasta salad	applesauce	juice

8 possibilities

- Compare your calculations with your partner. Explain whether or not you solved the problem the same way.
Answers will vary.
We both took the number of outcomes for each step and multiplied together to find the total number of outcomes. $2 \times 2 \times 2 = 8$ outcomes.



14.7.2 Collaboration: Making Predictions with Punnett Squares

Math can be used to help make sense of scientific processes. To study genetics, scientists use Punnett Squares. Punnett Squares can be used to determine what traits parents can pass on to their offspring and how likely it is that they pass on certain traits. Consider a litter of puppies in which the father of the dogs has a short tail, and his genetic code for that is "Tt." The mother of the litter also has a short tail with the same genetic code "Tt." Here is how a Punnett Square can be used to determine the sample space for tails of the puppies in the litter.

Punnett Square 1		Father's Code	
		T	t
Mother's Code	T	TT	Tt
	t	Tt	tt

- Punnett Square 1 shows that the puppies in the litter can either have a long tail (tt) or a short tail (Tt or TT). The Punnett Square can be used to make predictions about the litters of puppies from these parents to find how likely it is for the puppies in the litter to have certain traits.
 - Find the theoretical probability that a puppy from this litter will have a long tail (tt).

$$P(\text{long tail-tt}) = \frac{\text{the number of combinations that result in a long tail}}{\text{the number of total combinations}}$$

$$P(\text{long tail-tt}) = \frac{1}{4} = \frac{25}{100} = 0.25 = 25\%$$

- Find the theoretical probability that a puppy from this litter will have a short tail (TT or Tt).
- $$\frac{3}{4} = 0.75 = 75\%$$
- Use one of these probabilities to predict how many puppies in a litter of 8 would have long tails.

According to the Punnett Square, there is a $\frac{1}{4}$ chance these puppies will have a long tail. That means in a litter of 4 puppies, one is expected to have a long tail. If the litter has double the puppies, or eight puppies, equivalent fractions can be used to calculate the number of puppies who are predicted to have a long tail.

For example, $\frac{1}{4} = \frac{2}{8}$, **2** dogs are expected to have a long tail from the litter of 8.

- Throughout their lives, these two dogs were parents to 4 litters, a total of 28 dogs. Of these 28 dogs, how many are expected to have short tails?
21 of the 28 dogs are expected to be born with a short tail.
- Complete Punnett Square 2 to make a different kind of prediction. This Punnett Square shows the possible traits of puppies whose father's tail code is "tt," meaning he has a long tail, and whose mother's tail code is "Tt," which results in a short tail.

Punnett Square 2		Father's Code	
		t	t
Mother's Code	T	Tt	Tt
	t	tt	tt

14.7.2 COLLABORATION: MAKING PREDICTIONS WITH PUNNETT SQUARES

DIFFERENTIATE INSTRUCTION

This Collaboration includes a range of vocabulary words that can be used interchangeably. Prior to completing the Collaboration, clarify meanings. Consider asking students which pairs of words are synonyms, such as: The word: is a synonym for: probability expectation offspring likelihood prediction expectation offspring likelihood litter expectation offspring likelihood !!!

DIFFERENTIATE INSTRUCTION

Consider turning this Collaboration into a jigsaw activity in which groups of students become experts on one of the questions and then "teach" the class. This is an opportunity to differentiate based on observations from previous lessons through strategic grouping and question assignment.

TEACHER MOVES

Take Turns

Consider pairing this Collaboration with a Take Turns routine. Students in each group take turns completing the work of the problem and explaining their strategies and answers to each other. This routine encourages students to engage in mathematical discourse, which can help solidify understanding as students use Punnett Squares to determine probability.

14.7.2 COLLABORATION: MAKING PREDICTIONS WITH PUNNETT SQUARES (CONTINUED)

TEACHER MOVES

Students may struggle with question 4c, determining the total number of puppies. Encourage students to use what they know from questions 4a and 4b to help them find the answer for question 4c. Consider additional scaffolding through questioning:

- 12 puppies have short tails, what percentage of the total does 12 represent? (50%)
- If 12 represents half the total, how can you find the total? (by multiplying 12 by 2)

ENRICHMENT

This question asks students to find the likelihood of an event that is impossible. Pose questions to check student understanding of 0% probability, such as:

- What fur pattern is possible according to Punnett Square 3?
- What fur pattern is impossible according to Punnett Square 3?
- What probability is used to represent an impossible event?
- How can the probability you found for splotchy fur be interpreted?

- a. What is the likelihood that these offspring will have short tails? Write the answer in decimal form.
0.5

- b. What is the likelihood that these offspring will not have short tails? Write the answer in decimal form.
0.5

- c. Suppose that over time these parents have 12 puppies with short tails. How many total dogs are expected for these parents to have had?
Use equivalent fractions to calculate how many total puppies these two parents had over time.

$$\frac{1}{2} = \frac{12}{24}, \text{ 24 dogs total}$$

5. Punnett Squares can also be used to make predictions about fur patterns in a litter of kittens. The code for solid fur is "ff." Cats with striped fur have the code "Ff" (one of each letter), and cats with splotchy fur have the code "FF."

- a. Use Punnett Square 3 below to show the possible traits a litter of cats could have if their father has solid fur (ff) and their mother has striped fur (Ff).

Punnett Square 3		Father's Code	
		f	f
Mother's Code	F	Ff	Ff
	f	ff	ff

- b. What is the likelihood that these two cats will have offspring with splotchy fur (FF)?
0%

- c. At a cat sanctuary, 150 cats are born to one parent with solid fur (ff) and one parent with striped fur (Ff). Explain how many of these cats you would expect to have striped fur.
75 cats

6. Suppose a mother cat with striped fur (Ff) and a father cat with striped fur (Ff) have 36 kittens with striped fur over the years.

Punnett Square 4		Father's Code	
Mother's Code	F	FF	Ff
	f	Ff	ff

- Complete Punnett Square 4 to determine all possible fur patterns for their offspring.
- Predict how many total kittens these parents had.
72 total cats
- Predict the number of kittens these parents had with solid fur.
18 cats
- How many of these kittens are expected to have splotchy fur?
18 cats

14.7.2 COLLABORATION: MAKING PREDICTIONS WITH PUNNETT SQUARES (CONTINUED)

DIFFERENTIATE INSTRUCTION

Encourage students to label what each combination (Ff, FF, ff) means on the Punnett Square to provide an additional visual entry point.

TEACHER MOVES

Students may struggle to find the number of kittens for questions 6a-6c. Encourage students to use the Punnett Square to determine the percentages of each of the four options and then determine the numbers from there. Point out that 36 represents the number of kittens with striped fur and guide them towards determining the percentage of the whole that represents before they complete this section.

14.7.3 YOUR TURN!

DIFFERENTIATE INSTRUCTION

This section provides a visual entry point through the use of the diagram. Encourage students who are struggling to work independently to utilize the diagram as a jumping-off point for determining their answers and predictions.

QUESTIONING

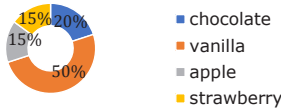
After completing question 1c, challenge students to determine if a pattern exists among the predictions. Consider providing a table for students to use to organize their findings, such as: Alex's Bakery Chocolate 20% 107 Vanilla 50% Apple 15% Strawberry 15% Total 100% 1,000 4,000
!!!



14.7.3 Your Turn!

- At Alex's Bakery, the owners keep detailed records about their sales. The graph below shows the probability that each flavor of doughnut will be purchased based on first quarter sales.

Probability Each Flavor is Sold



- If a total of 1,000 doughnuts are sold, predict how many of those are apple flavored.
150 doughnuts
 - Suppose that 107 chocolate doughnuts are sold in a day. Predict how many total doughnuts were sold this same day.
535 doughnuts
 - In one year, Alex's Bakery sells about 4,000 doughnuts. Estimate how many of those doughnuts are vanilla flavored.
2,000 doughnuts
- Darrielle picks 10 shoes out of her closet without looking. One fifth of them are blue. If there are 20 pairs of shoes in her closet, predict how many of those pairs are not blue.
Roughly 16 pairs are not blue.

14.7.4 WRAP-UP: TICKET OUT THE DOOR

**14.7.4 Wrap-Up: Ticket Out the Door**

1. A 50-card deck contains one card for each of the U.S. states. Four of the U.S. states begin with the letter "A". If Grant randomly selects a card, with replacement, 200 times, predict the number of repetitions of the experiment that will result in a state that begins with the letter "A."

Roughly 16 of the 200 will result with the starting letter "A".